

Team Control Number

14749

Problem Chosen

C

2023

MidMCM

Summary Sheet

1. Summary

The problem that our team chose was problem C. In this problem, we are required to estimate how many medals and flags will be needed for the 2024 Paris Olympics and determine how many medals and flags should be ordered in preparation for the event while justifying our decisions.

The problem states that the 2024 Paris Olympics will involve 203 eligible National Olympic Committees (NOCs), each represented by their own unique flag. There will be a total of 329 medal events that fall under the umbrella of 40 sports. Gold, silver, and bronze medals will be awarded to individual athletes or teams based on performance. The flags of the top three countries in each medal event will be displayed during the respective award ceremonies.

The Paris 2024 Olympics will take place across 37 venues, including locations all around Paris, as well as some overseas venues in Tahiti. These venues have been strategically selected to accommodate the diverse range of sporting events. There will be up to 203 countries competing under their own unique flag in the 2024 Paris Olympics, and there are 37 venues that the Olympics will be held across. This problem requires us to focus on one of three: the La Défense Arena, the Bercy Arena, the Stade de France. The problem requires us to create a schedule for the medal ceremonies in one of the individual venues listed above.

Under the guarantee that we will certainly not be missing any medals or flags to avoid embarrassment, we should try to save as much money and resources as possible. To ensure efficient planning and avoid unnecessary expenses, the International Olympic Committee (IOC) has sought assistance from MidMCM Olympic Consultants. The goal is to determine the quantity of medals and flags required for all ceremonies and create a model to do so (that can also be applied for other events). Accurate mathematical models and analyses will be provided to help the IOC make informed decisions. In the end, we had to share our solution by writing a one-to-two-page letter to the International Olympic Committee (IOC) to present, justify and explain our model.



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2. Introduction

2.1 Background

The Summer Olympic Games, held once every four years, is one of the most prestigious miscellaneous sporting events globally. The Olympics promotes the convergence of nations through sports, wherein each participating country sends their most prominent athletes to compete in a diverse array of events. The organizational, managerial, and regulatory aspects are entrusted to the International Olympic Committee (IOC).^[1] The Olympics originated in ancient Greece, and the ancient Olympic Games were held in the city-state Olympia from 776 BCE to 393 CE, making the games one of the oldest documented sporting events in history.

The Olympics, with its long history and worldwide acclaim, captivates the attention of an immense global audience. Data from Statista reveals that the average aggregate viewership for the preceding four Olympic Games approximated a staggering 3.34 billion individuals^[4]. But the significance of the Olympics is substantially more than simply a source of entertainment. It functions as a platform wherein athletes can exhibit their aptitude and commitment on an international stage. Furthermore, the Games provide athletes with an opportunity to congregate under their respective flags and evokes a sense of pride and patriotism.

This problem is important because if we do not calculate the number of medals needed correctly, then the IOC will face embarrassment and questions about its credibility, which we need to prevent. The Olympics also loses its prestige as one of the largest global sporting events, and the host country (in this case, Paris) is likely to have to take on the burden of being “irresponsible”.

2.2 Restatement of Problem

In this problem, we were tasked with determining the appropriate quantity of medals and flags that the IOC should order for all the venues and award ceremonies, ensuring that there are no shortages. However, we should not order too many as it would be a waste of money and resources. We must consider all situations within the competition that could happen, from ties in ranking to the number of representatives that a country sends. With consideration of these scenarios, we are asked to create a model of how many flags and medals we will require and derive a formula that can be applied to other venues to calculate flags and medal requirements. Then, we should create a schedule for the award ceremonies, and finally, write a letter to the IOC regarding our model and schedule.

3. Inquiry

3.1 Questions



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A lot of uncertainties arise from this problem, and it is crucial to provide comprehensive answers to ensure a thorough understanding. The following is a compiled list of the key questions that we considered while solving this problem:

- How many events will require medals in a single arena?
- What are the potential special situations that could impact our results and calculations?
- In the case of unforeseen circumstances like technical or mechanical issues, how can we adjust the schedule and allocate resources accordingly?
- Which historical data can be applied to future scenarios, and how can we demonstrate the sustainability of these trends?
- What is the maximum and minimum number of medals that will be required?
- What flag requirements does our selected venue have, and what is the optimal quantity for each type of flag?
- How can we demonstrate that our schedule is the most efficient and effective?
- How can we justify the costs and measures taken as precautionary measures?
- What should be done with any remaining or spare medals and flags?
- How can we calculate the final cost of our model?
- Which countries will participate, and in which events will they compete?
- How does historical data translate to future Olympic data?
- How can we introduce constraints in the schedule to make it more athlete-friendly?
- How can we allocate sufficient time between events to accommodate unexpected scenarios?

Answering these questions will guide us through our answer to this problem and help us create a more well-rounded answer. These questions also help our group make reasonable decisions and conclusions while coming up with a solution.

3.2 Assumptions

Below is a list of assumptions that must be clarified before continuing with the paper. These assumptions, all coming with their respective justifications, will help us answer the above questions, make future claims, and explain some of our decisions.

Assumption 1: The countries participating and medals necessary are independent factors

Justification 1: No matter how many countries participate in each event, there still will be a first, second, and third place. Thus, if there are at least three countries participating (which there are for every event), the event will need three or more medals. Each country's participation will only be needed in determining the flags required and the schedule for a venue.



Assumption 2: Most countries will not be competing in every event available, meaning some venues will not require flags of certain countries.

Justification 2: There are two main reasons for this. First, all Olympic events require a qualification process – meaning that the countries who do not do well historically in this event cannot participate in it. Secondly, most countries do not have the resources and capacity to consistently maintain 329 sport teams at a highly competitive, Olympic level. Since flags are only required for the top 3 placing countries, we can also assume, then, that a venue will not require flags of every participating country.

Assumption 3: Countries have the incentive to send more than one representative.

Justification 3: If there are multiple representatives, there is a higher chance of winning a medal for the country. Furthermore, it is very unlikely for an individual representative from a country to make it to the Olympics, as it takes plenty of resources to do so which usually means a resource-heavy system that can create more athletes.

Assumption 4: The composition of the medal would most likely be the same as the ones from earlier games.

Justification 4: The medal's composition was decided as such (specific) because it was the most cost-efficient without sacrificing its quality and value. The Olympics have no reason to change the composition of the medal and have not altered it much (citation needed) in the past few Games. As such, our calculations for the price of medals can be made using similar compositions as previous games.

Assumption 5: Not all countries will accept the invitation to the Olympics, but most will.

Justification 5: The Olympic Games are the largest sporting event globally, and joining it brings the country lots of benefits. For instance, its country benefits from pride, media exposure, and if they win events.

Assumption 6: The consequences of having an extra flag or medal are less severe than the consequences of missing any flags or medals.

Justification 6: There is less negative impact of having more flags than having too few. This is because there can always be other uses. For instance, if we have extra flags, we can still store them away and use them for future events. In the worst case, flags are cheap in monetary value – at the end of the day, it's not as if we are wasting large sums of money by ordering extra flags. In terms of medals, these medals can be used for other events in the Games or re-used for other projects that need similar materials and compositions. On the other hand, if we are missing any,



it would cause embarrassment, as well as be a sign of undermining and disrespecting a country, tarnishing the Olympics' credibility and reputation as a top global sporting event. This is also stated on the packet.

Assumption 7: No two events will be held at the same time, but award ceremonies of one event may be held at the same time another event takes place.

Justification 7: There is simply not enough room in the arena to host more than one event at once. However, as the award ceremonies do not use the pool, they can happen at the same time. For Water Polo events especially, they take much longer time than swimming events and so this will be needed in some cases, if some events go severely over time. This is not to say that we will try to squash award ceremonies with other events together; That would be worse for us, as there would be a conflict in broadcasting times and audience, which influences viewership. This assumption will come into play if unexpected events suddenly arise.

4. Developing Our Models

4.1 Consideration of Different Scenarios

Many events could happen that can lead to vastly different results, and it is our burden to ensure that we are prepared for these situations. In this section, we will go over major scenarios that could result in more/less medals and/or flags being required.

While there are undoubtedly more scenarios that can happen in the Olympics, we will ignore the scenarios that make no difference in terms of medal and flag preparations from the perspective of the IOC. Factors like whether an individual athlete participates in multiple Swimming events are important to organize the Olympics, but not important in scheduling and creating flag and medal models.

- a. Swimming events could have draws, resulting in more demand for a single type of medal. As such, it is justified for us to prepare one extra of each type of medal for all events. According to Swimming World Magazine, there have only been three gold-medal ties in the history of Olympic swimming, and they all happened in sprint-freestyle events. The first tie happened in 1984, where Americans Nancy Hogshead and Carrie Steinseifer tied the game in the 100m freestyle at the Los Angeles Olympics. 16 years later at the Sydney Olympics, Gary Hall Jr. And Anthony Ervin each earned a gold medal in the 50m freestyle event. The last draw happened at the 2016 Rio de Janeiro Olympics, where Simone Manuel and Penny Oleksiak posted identical times in the 100m freestyle. If draws were to happen within the top three, then an extra medal would be required for all the athletes who tie. ^[5]However, these scenarios are extremely rare since it only happened 3 times in Olympic history, and all happened in the sprint freestyle event. Meanwhile, water polo is impossible to have a tie, because if the game is tied in regular game time, there will be a penalty shootout. Therefore, we do not have to prepare too many extra



medals for this case, but we cannot eliminate the possibility of a tie as they have happened before.

- b. In swimming, multiple athletes representing the same country can rank in the top three, which results in one country winning more than one medal in one event. In this case, there would be duplicate flags required for the award ceremonies, as multiple flags of the same country will be flown during the ceremony. Up to three flags could be needed if one country's athletes take first, second, and third place, especially for countries that have been historically dominant in a certain event.
- c. There are many examples of ways that a team could have their result disqualified after the competition and even the award ceremony. The most prominent is drug testing, as drug test results come out after the award ceremonies. Thus, if a country tests positive for drugs, extra medals will have to be provided to the new winners, as medals cannot be recycled from disqualified teams. However, flags will not be a part of this consideration, as the award ceremony will not be held again. For example, women's weightlifting athlete Christine Girard was awarded replacement medals in 2016 after her competitors in the 2012 and 2008 Olympics were caught using drugs.^[13]
- d. Medals and flags could be damaged, and as a result, require replacement. This situation is not very uncommon, as Olympic Medals are expected to be given at an extremely high quality as a symbol of winning in a globally large sporting event, and flags, as the representation of a country, have no room for error. As a result, even slight defects in medals or flags could call for replacements and extras. We need to prepare extra materials in the case that we have to make replacements.

4.2 Base Model for Medals

In this problem, we chose to work with the La Defense Arena, which hosts Water Polo and Swimming events.

First, we calculated the minimum number of medals that would be required, assuming nothing goes wrong. From now on, this will be referred to as the “base medals”. To calculate the number of base medals that will be needed for a certain arena, we first must find out how many medals each event will require. To find this, we will use the formula of:

$$E_x = C \times T$$

Where E_x is the number of medals required for event number x (as listed on the table), C is the number of categories within an event, and T is the team size of each event.

This formula will be used in the “Medals Required” column.

Table 1. The medals required for swimming events at the Paris Olympics, as found on the official Paris Olympics website

Event	Medals required*	Which categories?
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50m Freestyle (1)	2	Men's, Women's
100m Freestyle (2)	2	Men's, Women's
200m Freestyle (3)	2	Men's, Women's
400m Freestyle (4)	2	Men's, Women's
800m Freestyle (5)	2	Men's, Women's
1500m Freestyle (6)	2	Men's, Women's
100m Backstroke (7)	2	Men's, Women's
200m Backstroke (8)	2	Men's, Women's
100m Breaststroke (9)	2	Men's, Women's
200m Breaststroke (10)	2	Men's, Women's
100m Butterfly (11)	2	Men's, Women's
200m Butterfly (12)	2	Men's, Women's
200m Individual Medley (13)	2	Men's, Women's
400m Individual Medley (14)	2	Men's, Women's
4x100m Freestyle Relay (15)	8	Men's, Women's
4x200m Freestyle Relay (16)	8	Men's, Women's
4x100m Medley Relay (17)	12	Men's, Women's, Mixed
Total	56	n/a
Source: Paris Olympics Website ^[2]		

*Numbers refer to how many of each medal type in this event - for instance, 2 means 2 gold, 2 silver, 2 bronze.

For water polo, there is no need for a table as there are no events within. There are only two categories: male and female. Each water polo team consists of 13 players. Thus, using $E_x = C \times T$, can find that all water polo events would require a total of 26 gold, 26 silver, and 26 bronze medals.

This model and equation is applicable to other events and venues too. The logic remains the same for all events, as each event has divisions and team size which directly determine the medals needed for it.

We can use this formula (listed below) to determine the base medals (B) required for a given venue:

$$B = E_1 + E_2 + E_3 \dots$$

Therefore, assuming everything goes to plan, we would need 82 gold, 82 silver, and 82 bronze medals for the La Défense Arena.

4.3 Consideration of scenarios for medals

We previously produced four possible scenarios, three of which could impact the total medal count required.



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The first scenario of a tie is very unlikely as we have previously explained. However, it cannot be completely taken out of consideration as it has happened three times historically. As such, it is justified for us to prepare one extra of each type of medal for the venue. The reason we do not expect draws to happen for relay events is because it has never happened historically and has much less probability of happening in a relay event than a single-swimmer event, as with four people, there is more room for variation compared to one. Furthermore, the relay events are all of 100 meters (individually) or more, but the only draws to happen in Olympic history are of sprint (50m) freestyle events. For other venues, we can take the historical data of events located there and take the ratio of the number of draws in recent years over the number of total events held (which we will call R). Then, we will multiply the number of base medals by R and round to the nearest whole to find the extra medals we will prepare in the event of a draw.

The second scenario would only impact the flag model, which will be covered later. It will not have any impact on the medal model.

The third scenario is disqualification after award ceremonies, which calls for replacement medals to be awarded to the new winners. We must consider this now because we never know when these replacement medals will be required – it could be anywhere from a few days after the event to years later. We will have enough medals to cover if one water polo team is disqualified and if one relay team is disqualified, for a total of 17 medals in this section. This will be combined with scenario 4, aspect 2. For other venues, the number of medals they prepare should be the sum of the two largest values of E_x . We will call this sum the variable F .

The fourth scenario of defects in medals has two aspects. The first aspect is if the medal is found defected before awarded to an athlete. In this case, the IOC will only have to switch out a medal to distribute. We will prepare four extra medals for this, as this should not be common, especially given the brief time from creation to distribution. This consideration is listed as “Extra Medals” on the table and can be calculated as 4.5% of the base models for other venues.

The second aspect is whether the medal must be replaced after it is handed out. An example would be the 2016 Olympic Games, where the Rio Games spokesperson Mario Andrada stated that, “officials have noted problems with the covering on 6 to 7 percent of the medals”^[3]. Of course, we would not expect as many medals to be defects this time as this was an outlier, but we chose to prepare five percent of the total medals for post-distribution defects, which is 4 medals in this case. This combined with scenario 3 forms the section of “future compensation”, with 21 medals. If more medal replacements are needed, they can be made on the spot with the medal replacement funds taken as a part of the medal replacement process. With all the above considerations, we can then conclude the total amount of medals needed.

Table 2. Medals needed for La Défense Arena with consideration of scenarios

Event	Medals required*	Which categories?
50m Freestyle	2	Men's, Women's
100m Freestyle	2	Men's, Women's
200m Freestyle	2	Men's, Women's



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400m Freestyle	2	Men's, Women's
800m Freestyle	2	Men's, Women's
1500m Freestyle	2	Men's, Women's
100m Backstroke	2	Men's, Women's
200m Backstroke	2	Men's, Women's
100m Breaststroke	2	Men's, Women's
200m Breaststroke	2	Men's, Women's
100m Butterfly	2	Men's, Women's
200m Butterfly	2	Men's, Women's
200m Individual Medley	2	Men's, Women's
400m Individual Medley	2	Men's, Women's
4x100m Freestyle Relay	8	Men's, Women's
4x200m Freestyle Relay	8	Men's, Women's
4x100m Medley Relay	12	Men's, Women's, Mixed
Water Polo	26	Men's, Women's
Event Medals Total	82	n/a
Consideration of possible ties	1	n/a
Consideration of future compensation	21	n/a
Extra Medals	4	n/a
Additional Medals Total	26	n/a
Total Medals Required	108	Event Medals, Additional Medals

But, what will we do with the medals if they are left over? In other words, why is it justified for us to be spending more money and resources in anticipation for relatively rare scenarios, and what will we do with the leftovers?

We've already explained the importance of ensuring enough resources in the Assumptions section. But embarrassment factors aside, there are three main places these spare medals can go to:

First, the medals can be stored in case there is more need for future medal compensation, usually found in scenario 3 (disqualifications in the future) or scenario 4 (requirement for replacement medals) happen in the future. Since the replacement medals should have the same design as those given in the Olympics, having extra medals will mean that there won't have to be medals made whenever they are needed however long in the future, and will already be ready.

Second, we have seen that the medal composition stays consistent through different Olympic events. This means that the material of these medals can be reused for future Olympic Games medals too, especially given that the quality of the medal won't deteriorate in four years (if kept under good condition).



Third, these spare medals can be used in the event of any unexpected events that take place in other Olympic arenas at the event. Since the design remains consistent throughout the whole Paris Olympics, spare medals from one venue can be given to another venue that is lacking medals.

For all other venues, a similar approach can be taken to determine how many medals will be needed there. Our formula for determining the total number of medals (M) needed at a given venue is:

$$M = B + B(R) + F + 9.5\%(B)^*$$

*The 9.5% comes from the sum of the two events considered in scenario four, which require 4.5% and 5% of B respectively.

4.4 Medal Prices

In this section, we will look through the prices to produce each type of medal, then multiply them by 108 (as we will need 108 of each medal type), and finally add them all together to find the total cost of medals. All material prices in this section are as of November 14, 2023.

First, we will calculate the price of gold medals. Gold medals are required to have at least 92.5% silver, and usually weigh 556 grams^{[9][10]}. Gold medals have a gold plating on the outside consisting of at least 6 grams of pure gold. Thus, the cost of gold in each medal can be found by $6 \text{ grams} \times \$62.57/\text{g}^* = \375.42

*price of gold (per gram)^[11]

The cost of silver in each medal can be calculated by $(\text{Total weight} - \text{Weight of gold}) \times \$0.71/\text{g}$ (price of silver per gram). The weight of silver in each gold medal is the total weight subtracted by the weight of gold, which in this case, is 556 grams - 6 grams, or 550 grams per medal. Thus, the cost of silver in each gold medal is $550 \text{ grams} \times \$0.71/\text{g} = \$390.50$.

Then, we can calculate the total cost per gold medal by adding these two together, to get a total of \$765.92 per gold medal. Therefore, the cost of 108 gold medals (the amount required for the arena) would be $(108 \times \$765.92)$, or \$82,719.36 in total.

Now, we will look into silver medals. Silver medals are made entirely of pure silver. Therefore, the cost of a whole silver medal would be simply the total weight $(550\text{g})^{[9]} \times \$0.71/\text{g}$, which would be \$390.50. Therefore, the cost of 108 silver medals is $(108 \times \$390.50)$, or \$42,084.00 in total.

Moving on, we will investigate bronze medals. Bronze medals are composed of 95% copper and 5% zinc. Currently, zinc costs approximately \$0.0026/g, and copper costs \$0.23/g^[12]. The weight of copper in each bronze medal is the total weight $(450\text{g}) \times 0.95$, which is 427.5 grams. Multiply that by \$0.23/g, and we get that the cost of copper in each bronze medal is \$98.33. The



weight of zinc in each bronze medal is 450 grams $\times$ 0.05, or 22.5 grams, which costs \$0.06. Thus, the total cost per bronze medal is \$98.39. (\$98.33 + \$0.06), and the cost of 108 bronze medals is $(108 \times \$98.39)$, or \$10,627.12 in total.

Lastly, each medal requires a ribbon. There is no way to provide a specific cost to the ribbon, but since its material (polyester) is extremely cheap, we can use a general estimate of \$0.05 per ribbon. Therefore, the cost of ribbons for all medals would be $(108 \text{ gold medals} + 108 \text{ silver medals} + 108 \text{ bronze medals}) \times \$0.05 = \$16.20$.

With all the above calculations \$82,719.36 (gold) + \$42,084.00 (silver) + \$10,627.12 (bronze) + \$16.20 (ribbons) = \$135,446.68.

4.5 Model for Flags & Justification.

Our model consists of getting two flags in our water sport system for every country that we have reason to believe will send more than one representative in one of the events. If there is a need for a third (which is a rather uncommon scenario), we will reuse the flags from the opening ceremony.

Our alternative of re-using flags from the Opening Ceremony is justified and feasible, as the Opening ceremony will be held at Seine , which is just over a 1-hour drive from the La Défense Arena.

Here is historical data, regarding the most recent swimming event winners, by country, in the 2020 Tokyo Olympic Games. Countries who did not win any medals were not included on this table.

Table 3. 2020 Tokyo Olympic Games swimming award winners

		2020					
Rank	Country	Gold	Silver	Bronze	Total	Medals Ratio*	Flags needed
1	United States	11	10	9	30	0.27	2
2	Australia	9	3	9	21	0.19	2
3	Great Britain	4	3	1	8	0.07	2
4	China	3	2	1	6	0.05	2
5	ROC	2	2	1	5	0.05	2
6	Japan*	2	1	0	3	0.03	2
7	Canada	1	3	2	6	0.05	2
8	Hungary	1	2	0	3	0.03	2



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9	South Africa	1	1	0	2	0.02	2
10	Brazil	1	0	2	3	0.03	2
10	Germany	1	0	2	3	0.03	2
12	Tunisia	1	0	0	1	0.01	2
13	Netherlands	0	3	0	3	0.03	2
14	Italy	0	2	5	7	0.06	2
15	Hong Kong	0	2	0	2	0.02	2
16	Ukraine	0	1	1	2	0.02	2
17	France	0	1	0	1	0.01	2
17	Sweden	0	1	0	1	0.01	2
19	Switzerland	0	0	2	2	0.02	2
20	Denmark	0	0	1	1	0.01	2
20	Finland	0	0	1	1	0.01	2
Totals (21 entries)		37	37	37	111	-	42

Table 4. Water Polo winners of Tokyo Olympics ^[8]

Rank	NOC	Gold	Silver	Bronze	Total
1	 Serbia	1	0	0	1
	 United States	1	0	0	1
3	 Greece	0	1	0	1
	 Spain	0	1	0	1
5	 Hungary	0	0	2	2
Totals (5 entries)		2	2	2	6

*Medals Ratio refers to Medals earned: Medals possible (total available medals), rounded to the nearest hundredth.

Since it was difficult to find the maximum number of flags active at one time, we would need to estimate to need 3 flags for every country. However, given that multiple sports might be hosted at the times, the number of flags that a country will need is the number of medals a country wins in the previous Olympics, totaled up is higher (Bronze, Gold and Silver), over one third number of competitions, rounded up + 1. Say Argentina wins 87 Bronze medals, 98 Silver medals and 57 Gold medals, and there are 300 rounds, we would need 1 flag for the country. This venue in total has 37 “medal-giving” ceremonies, and assuming the averages are consistent,



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we would use a data management system and quickly figure out the number of flags a country needs.

Table 3 (above) shows that we would need three times the flags for the US, Australia, Great Britain, China, ROC, Japan, Canada, Hungary, South Africa, and double the flags for Brazil, Germany, Tunisia, Netherlands, Italy, Hong Kong, Ukraine, France, Sweden, Switzerland, Denmark and Finland for swimming competitions, as per the results of the previous Olympics.

Similarly, we decided that the probabilities of having one extra flag and one wasted flag are equally likely, because if the number of necessary flags for a country is an unknown constant between 0 and 3. By the chart below, we would have an equal chance to get it over than to get it under.

Table 5. Probability chart for flags

	0	1	2	3
0		Under	Under	Under
1	Over		Under	Under
2	Over	Over		Under
3	Over	Over	Over	

X-axis = Number of flags needed

Y-axis = Number of flags provided

Because the consequences for assuming the amount of flags required under (excessive humiliation by the media, harming credibility of the Olympics, etc.) are way larger in significance compared to the consequences of going over (Wasted flags, which are cheap in the first place), we believe that all countries should be assigned 2 flags for their swimming part, because the chances of winning all 3 medals are very low (only 5 instances), and by the chart, we would have saved the resources of a flag for a low chance of humiliation. Even if such an event were to happen, we could borrow a flag from the opening ceremony or a neighboring stadium for a short time period.

Water polo is much simpler, as there is no possibility for a country to win more than one place within the top three. Thus, there will need to be just one flag for each participating country. There will be 12 Men's Water Polo teams and 10 Women's Water Polo teams.

Since the qualification process for the Olympics has still not been completed yet, we cannot be certain as to which countries would require flags to be created. However, a lot of countries that participate

To determine the flags that another venue would require, we would first look at whether there is a possibility for multiple countries to participate. If the event is like Water Polo in that there is no scenario which would require multiple flags, then the IOC simply needs to prepare one flag for each country. If there is room for multiple rankings within the top three, then the IOC should prepare two to three flags for each country, depending on whether there is a viable



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alternative to obtain another flag from an external source (e.g. neighboring facility) if a third flag is needed, or whether there is any country with more than two participants in the first place.

4.6 Costs & Cost Justification

Since every flag is mostly the cost of the fabric and dye, (With room for labor costs), we believe that every flag would cost at least \$300. That way, we would need $203 \times 2 \times 300 = \121800 dollars. In the past, the IOC never lost any necessary flags, and this is the “better safe than sorry” approach which was previously justified.

4.7 Generalization Considerations

This model is extremely generalized, because the main constraints are that no two events will be hosted at the same time, so if the stars align, and two competitions were hosted at the same time, both have 1st, 2nd and 3rd won by the US, that will make us need 6 flags. Therefore, this constraint's purpose is to limit the maximum number of flags used at one time.

4.8 Schedule for Award Ceremonies

The Paris Olympics will be hosted across 17 days, from July 26, 2024 to August 11, 2024. Excluding the first day as it is the opening ceremony, that leaves us with a timespan of 16 days to fit all events and award ceremonies for Water Polo and Swimming.

We can find the schedule for the Swimming and Water Polo events (which has already been made) on the NBC^{[6][7]}. From these, we can find out when each event ends, and when there would be time to hold an award ceremony for these events. The schedule is relatively vague, but we considered factors such as the distance of each swimming event and It is justified for us to go until 11:35 PM as there are no swimming events before 11:00 AM local time, which means they can still get the sleep they need. However, we will not go past this time as if we must go past 11:30, we would rather hold the ceremony the next morning (before all other events) to ensure the athletes get enough rest and can compete at a competitive level without being handicapped by our schedule. We will ensure that there are at least 11 hours and 30 minutes between two possible events that the athlete could possibly participate in. We hold award ceremonies at separate times because we want to minimize scenarios where award ceremonies are held during other events unless we have to, as that would mean less TV coverage. The more time the Olympics can be played on television, the more money the IOC earns.

Furthermore, we would like to guarantee at least 30 minutes of time between when the event and when its corresponding award ceremony takes place. This is so that the athletes can have sufficient time to organize and prepare themselves for the ceremony after the event ends, without having to hectically get ready for the award ceremony. We will also leave five minutes of time from when the last swimming events end to when the award ceremonies for other events are held, so that the venue organizers have time to set up and prepare for the ceremony.



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Lastly, we expect each award ceremony to take 15 minutes at most. Here is how we expect the award ceremonies to play out:

1. Athlete introduction & stepping onto podium – 2-3 Minutes
2. Medals, flowers, etc. (Award giving) - 2-3 Minutes
3. National Anthem & Flag Raising – 2-5 Minutes (depending on anthem length)
4. Photos – 1-2 Minutes
5. Extra Time Considerations (in case of unexpected scenarios) 1-2 Minutes

If we take the maximum time expected in these events and add them together, we get exactly 15 minutes. Therefore, 15 minutes will be the time we allocate for each award ceremony.

Table 6. Award Ceremony Schedule, in order of award ceremony time

Event	Time of Finals (Local Time)	Time of Award Ceremony (Local Time)
Men's 400m Freestyle	Jul. 27 20:50-21:10	Jul. 27 22:35-22:50
Women's 400m Freestyle	Jul. 27 21:10-21:30	Jul. 27 22:50-23:05
Women's 4x100 Freestyle Relay	Jul. 27 21:50-22:10	Jul. 27 23:05-23:20
Men's 4x100m Freestyle Relay	Jul. 27 22:10-22:30	Jul. 27 23:20-23:35
Men's 400m Medley	Jul. 28 20:30-20:50	Jul. 28 22:35-22:50
Women's 100m Butterfly	Jul. 28 20:50-21:10	Jul. 28 22:50-23:05
Men's 100m Breaststroke	Jul. 28 21:50-22:10	Jul. 28 23:05-23:20
Women's 400m Medley	Jul. 29 20:30-20:50	Jul. 29 22:35-22:50
Men's 200m Freestyle	Jul. 29 20:50-21:10	Jul. 29 22:50-23:05
Men's 100m Backstroke	Jul. 29 21:30-21:50	Jul. 29 23:05-23:20
Women's 100m Breaststroke	Jul. 29 21:50-22:10	Jul. 29 23:20-23:35
Women's 200m Freestyle	Jul. 29 22:10-22:30	Jul. 30 10:10-10:25
Women's 100m Backstroke	Jul. 30 21:00-21:15	Jul. 30 22:35-22:50
Men's 800m Freestyle	Jul. 30 21:15-21:40	Jul. 30 22:50-23:05
Men's 4x200m Freestyle Relay	Jul. 30 22:10-22:30	Jul. 30 23:05-23:20
Women's 100m Freestyle	Jul. 31 20:30-20:40	Jul. 31 22:35-22:50
Men's 200m Butterfly	Jul. 31 20:40-20:50	Jul. 31 22:50-23:05
Men's 200m Breaststroke	Jul. 31 20:50-21:00	Jul. 31 23:05-23:20
Women's 1500m Freestyle	Jul. 31 21:00-21:30	Jul. 31 23:20-23:35
Men's 100m Freestyle	Jul. 31 22:15-22:30	Aug. 1 10:10-10:25
Women's 200m Butterfly	Aug. 1 20:30-20:45	Aug. 1 22:35-22:50
Men's 200m Backstroke	Aug. 1 20:45-21:00	Aug. 1 22:50-23:05
Women's 200m Breaststroke	Aug. 1 21:15-21:30	Aug. 1 23:05-23:20
Women's 4x200m Freestyle Relay	Aug. 1 22:00-22:30	Aug. 1 23:20-23:35
Men's 50m Freestyle	Aug. 2 20:30-20:50	Aug. 2 22:35-22:50
Women's 200m Backstroke	Aug. 2 20:50-21:10	Aug. 2 22:50-23:05
Men's 200m Medley	Aug. 2 21:10-21:30	Aug. 2 23:05-23:20



Men's 100m Butterfly	Aug. 3 20:30-20:50	Aug. 3 22:35-22:50
Women's 200m Medley	Aug. 3 21:10-21:30	Aug. 3 22:50-23:05
Women's 800m Freestyle	Aug. 3 21:30-22:00	Aug. 3 23:05-23:20
4x100m Mixed Medley	Aug. 3 22:00-22:30	Aug. 3 23:20-23:35
Women's 50m Freestyle	Aug. 4 18:30-18:50	Aug. 4 20:35-20:50
Men's 1500m Freestyle	Aug. 4 18:50-19:30	Aug. 4 20:50-21:05
Men's 4x100m Medley Relay	Aug. 4 19:30-20:00	Aug. 4 21:05-21:20
Women's 4x100m Medley Relay	Aug. 4 20:00-20:30	Aug. 4 21:20-21:35
Women's Water Polo* **	Aug. 10 19:10-21:00	Aug. 10 21:30
Men's Water Polo*	Aug. 11 14:00-15:50	Aug. 11 16:30

*There are bronze medal games held for these events; However, they all take place before the final game, so they will not make any difference to the award ceremony planning.

**The starting time for this event is assumed based on the similar 1:50 time allocated for the Men's Water Polo final.

5. Conclusion

5.1 Analysis of Our Project

Our model ensures there will certainly be sufficient medals and flags for all events that will be held in the La Défense Arena. We have thoroughly considered lots of scenarios that could take place and ensured that there is no possibility for the IOC to end up embarrassed due to a lack of materials. Furthermore, we have justified and explained all of the decisions we made, including a step-by-step process as to how we came up with the numbers, formulas, and calculations we ended up with. Additionally, we have taken data from last year's Olympic Games and created assumptions and models from that evidence.

Mathematically, we have derived a formula for flags and models that can be applied to different venues and scenarios, as long as variables involving the number of events held in a venue and the average team size for a team in each event are found. We have also determined the probability of winning and data from last Olympics to calculate the number of flags required.

However, our model comes with some limitations too. For instance, we cannot predict what countries will accept the Olympic invitation, and so we must use more money and resources because we need to produce the flags of essentially all participating countries before the Olympics. Nor can we determine which countries will qualify for each event, as the qualification processes for these events have yet to be concluded. Furthermore, while we have looked at historical data and analyzed countries which have been historically dominant, we cannot eradicate the possibility of another historically nondominant country winning a medal. Especially given the nature of Swimming events, all it takes is one exceptionally talented athlete to win the whole event for the country.

Another limitation for this model is that lots of measures were taken to ensure nothing goes wrong, so is simply inevitable that some of it will not be used. We need to prepare more medals and flags than needed to prevent lacking them in the award ceremony, because we need



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to prevent IOC from embarrassment. While this is justified, it's inevitable that in hindsight, there will be more money spent than needed, and some of it will go to waste. However, at the very least, we have proposed actions to take regarding the leftover medals to try to minimize the waste.

5.2 Concluding Paragraph

Throughout this paper, we determined how many medals and flags the IOC should prepare for the La Défense Swimming and Water Polo Arena. While determining the flag and medal numbers that should be prepared, we made sure to take into consideration multiple scenarios that could impact the number of flags and models required and made adjustments accordingly. We came up with ways to deal with spare medals that justified these extra medals that were prepared, ensured these medals did not go to waste, and even benefit the IOC in the long term. In this process, we also derived a formula for medals and flags that can be applied to any other venues in the Olympic Games to find out how many medals and flags should be prepared there. Lastly, we created a schedule to run the events in the venue, ensuring our schedule is flexible enough to deal with unexpected delays, and considering what would be best to the athletes and organizers.

6. Letter

Dear Executive Board of the International Olympic Committee,

Our model should be the chosen model because it considers all realistic scenarios and makes according to adjustments to come up with the most cost and resource efficient model that will guarantee the IOC does not face embarrassment. Despite it is the minimalist approach in the list, we believe that the consequences are too large to face, and there are measures we can take to ensure the leftover medals do not go to waste. We will prepare a total of 108 gold, silver, and bronze medals for Water Polo and Swimming events in the La Défense Arena, 86 for the base scenario which assumes nothing goes wrong, and 22 in case of events that could alter the medal total. Each country that participates in Swimming will have 2 flags, and each country participating in Water Polo will have 1 flag (which can also be duplicate from Swimming flags). Lastly, our schedule involves multiple events put together over the span of the whole Olympics, ensuring that there is enough time between events to allow the athletes to transition and account for unexpected events.

There are a few reasons we have chosen to be rather safe than sorry in resource preparation. First, we want to prevent scenarios where you, and the Olympics Games, take a massive hit to their reputation globally. There has never been an account in the history of the Games where any medals or flags were missed, and we want to keep that up. Thus, within a reasonable amount, our number one priority is undoubtedly to ensure that no matter the scenario, we do not miss any flags or medals. In other words, this is our main priority, the prerequisite to all medal preparations. While our model may not be the cheapest medal out there, our safe approach can guarantee that essentially nothing would go wrong.



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Furthermore, we have measures that can be taken with the leftover materials to ensure they do not simply go to waste. In terms of flags, we have proven that they are cheap and efficient to make, so there is nothing wrong with preparing spares, especially given that we are only preparing one extra for each participating country. These flags can also be kept at the venues for memorial purposes or re-used for other events locally that require flags. In terms of medals, we have ways to put the leftovers to use. Extra medals can be saved if any other venues/events require more medals, or in case future compensation for any Olympic events (as all Paris Olympic medals have the same design) are required. Furthermore, we know that the medal composition for all Olympics are kept very similar as the requirements for medals do not change. Thus, the materials used in medals this year can be re-used for future medals, which means that the IOC's spendings on medals now are long-term investments, not overcautious wastes of money.

Our model also excels in that we have created formulas that can be applied to other Olympic venues to determine how many flags and medals would be needed there. The formula for medals (M) is $M = B + B(R) + F + 9.5\%(B)$, where B refers to the base medals required for the venue, or the number of medals required if no special scenarios were to take place and everything goes according to plan, and F refers to the sum of the medals required in the two events requiring the most medals. The formula for flags is split into two types. Assuming there is no possibility for multiple teams/athletes from the same country to place within the top three (e.g. Water Polo), the number of flags we will prepare is equivalent to the number of participating countries. If there is possibility for one country to win multiple places, then the IOC should have 2 flags ready per country. It is also worth noting that events held within the same/neighboring venues can share flags.

For this project, there were a lot of assumptions we needed to make to prepare enough medals for the Olympics. The most unpredictable factor of this project is regarding how draws, defections, and disqualifications will affect the number of medals needed at the award ceremony and flags, and how to create the most cost efficient and resource efficient method. Furthermore, we must ensure that in our schedule, there is sufficient time to account for these scenarios if they were to happen. Our model excels as we comprehensively considered all the scenarios that could happen, and we have made adjustments accordingly, justifying every decision we make. Our model will not be rendered invalid as soon as something goes wrong at the event, which is essentially inevitable as no event can be perfect. Instead, we thoroughly consider unexpected scenarios and build our model to withstand other events that we may not have predicted.

All in all, our model should be the chosen model for the following reasons. First, we considered all the scenarios that could possibly happen to cause a demand for extra medals and flags, and prepared extra medals and flags for these cases. We considered ties, compensations, and defections. This allows the IOC to have enough medals in the award ceremony and prevent embarrassment. Second, we made assumptions of how many flags are needed during the award ceremony, with clear explanation as to how we came up with those numbers and deriving a formula to calculate the medals and flags required for other venues at the event too. We considered that some countries are historically dominant at a specific event, and prepared extra



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flags for these countries since they may win two places in the first three. While there will most likely be leftovers at the end of the event, we have proposed actions that we can take to ensure those extras do not go to waste.

Sincerely,

Team # 14750

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